26900252 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFSC301 - FORENSIC TOXICOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Mention two examples of antidotes used in poisoning cases. [U] / [CO1]
2. A forensic sample is suspected to contain a phosphide compound. Which preliminary chemical test would you select and what is its purpose? [A] / [CO3]
3. What is meant by evidence material in food-poisoning investigation? [U] / [CO4]
4. Mention two signs of carbon monoxide poisoning. [U] / [CO1]
5. How can the Dithionite test be applied to Paraquat or Diquat examination? [A] / [CO3]
6. What are plant poisons? [U] / [CO1]
7. Differentiate between the analysis of ethyl alcohol in beverages and biological fluids. [An] / [CO2]
8. Explain the use of spectrophotometry in poison analysis. [U] / [CO3]

9. Why are instrumental techniques used for identification? [A] / [CO4]

10. Analyse how the route of ingestion can influence the absorption and distribution of a poison. [An] / [CO2, CO3]

11. How can HPLC be applied to poison detection? [U] / [CO3]

12. Explain one limitation of a toxicological method. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the principle and forensic application of the Conway diffusion method for the determination of ethyl alcohol. [U] / [CO1]

14. Analyse the role of preliminary colour tests in forensic toxicology and explain why their results require confirmation by more specific analytical techniques. [An] / [CO2]

15. Describe insect and animal toxins relevant to toxicological analysis. [U] / [CO4]

16. Analyse how clean-up and concentration procedures improve the reliability of the subsequent identification of drugs and poisons. [An] / [CO2]

17. Explain TLC examination of barbiturates and cannabinoids. [A] / [CO3]

18. Discuss the presentation of laboratory findings in a toxicological report. [A] / [CO4]

19. Analyse the factors that affect the extraction of acidic and basic drugs from viscera, blood, and urine. [An] / [CO2]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

20. Explain the general principles and procedures involved in the extraction and isolation of poisons from biological specimens. Discuss the factors affecting the recovery of poisons [K] / [CO1]

21. Analyse the differences between chemical and bacterial food poisoning with respect to their toxicological examination. [An] / [CO4]

[10x1 = 10]